

CURIONEST

Chapter 1 — Math & Measurement

Unit Pack · Measurement · Sig Figs · Density · Scientific Notation · Dimensional Analysis

WHAT'S INSIDE

- ✓ Complete concept notes with worked examples
- ✓ Practice sets in varied formats (short answer, MCQ, calculations)
- ✓ Key terms glossary + full answer key with worked solutions
- ✓ Original figures and Word-native equations — print-ready & editable

Made by a chemistry teacher — aligned to the standard US chemistry sequence.

In this unit, you will learn to:

1. Distinguish qualitative and quantitative observations
2. Identify SI base units and metric prefixes
3. Count significant figures and round calculations correctly
4. Calculate density using numerical set-ups
5. Convert between standard and scientific notation
6. Convert units using dimensional analysis

Unit Vocabulary — Write each definition in your own words

Word	Definition (your own words)
accuracy	
precision	
conversion factor	
density	
mass	
qualitative observation	
quantitative observation	
scientific notation	
significant figures	
unit	

LESSON 1: MEASUREMENT

Objective:

- Distinguish qualitative from quantitative observations
- Identify SI base units and metric prefixes

Chemistry is the study of matter and its changes. Matter can be described through its properties using two kinds of observations. Qualitative observations are descriptive and non-numerical (color, odor, texture). Quantitative observations are in the form of numbers and units (mass 5 g, temperature 371 K).

Every measurement has three parts: a number (magnitude), a unit (standard of comparison), and an indication of uncertainty. The SI system uses base units: meter (length), kilogram (mass), second (time), kelvin (temperature). Metric prefixes change the size of a unit by powers of ten: kilo (k, 1000×), centi (c, 1/100), milli (m, 1/1000), micro (μ, 1/1,000,000).

Worked Example

Classify each observation as qualitative or quantitative: (a) the metal is silver-colored; (b) it melts at 371 K; (c) its density is 0.97 g/cm³.

Step 1: Silver-colored describes appearance without numbers → qualitative.

Step 2: 371 K is a number with a unit → quantitative.

Step 3: 0.97 g/cm³ is a number with units → quantitative.

Answer: (a) qualitative, (b) quantitative, (c) quantitative.

Check Your Understanding

1. Which is a QUALITATIVE observation?

- (A) The liquid is clear
- (B) The mass is 45.8 g
- (C) The temperature is 298 K
- (D) The volume is 25.0 mL

Answer: _____

2. Which SI base unit measures temperature?

- (A) Celsius
- (B) Kelvin
- (C) Fahrenheit
- (D) Joule

Answer: _____

3. The prefix 'kilo' means:

- (A) $1/100$
- (B) $1/1000$
- (C) $1000\times$
- (D) $1/1,000,000$

Answer: _____

4. 1 centimeter is what fraction of a meter?

- (A) $1/10$
- (B) $1/100$
- (C) $1/1000$
- (D) $1/1,000,000$

Answer: _____

LESSON 2: SIGNIFICANT FIGURES

Objective:

- Count significant figures in measurements
- Round calculations to the correct number of sig figs

Significant figures (sig figs) are the digits actually recorded by a measuring device. RULES FOR COUNTING SIG FIGS: (1) non-zero digits are always significant; (2) leading zeros are NOT significant — 0.003080 has 4 sig figs; (3) captive zeros between non-zeros ARE significant — 1005 has 4; (4) trailing zeros are significant only if a decimal point is present — 2500 has 2, 2500. has 4.

RULES FOR CALCULATIONS: For addition and subtraction, round to the fewest decimal places. For multiplication and division, round to the fewest significant figures. NEVER be more precise than the least precise measurement.

Worked Example

How many sig figs are in 0.003080? Then compute 12.55×0.25 and report correctly.

Step 1: Count sig figs in 0.003080: leading zeros do NOT count; 3080 are significant (trailing zero counts, decimal present).

$$0.003080 \rightarrow 4 \text{ sigfigs}$$

Step 2: For multiplication, round to the fewest sig figs of any factor. 0.25 has 2 sig figs:

$$12.55 \times 0.25 = 3.1375 \rightarrow 3.1 \text{ (2 sigfigs)}$$

Answer: 0.003080 has 4 sig figs; the product is 3.1.

Check Your Understanding

1. How many sig figs are in 0.003080?

- (A) 3
- (B) 4
- (C) 5
- (D) 6

Answer: _____

2. How many sig figs are in 2500 (no decimal)?

- (A) 2
- (B) 3
- (C) 4
- (D) Cannot determine

Answer: _____

3. Compute 2.5×3.1416 with correct sig figs:

- (A) 7.854
- (B) 7.85
- (C) 7.9

(D) 8

Answer: _____

4. For addition/subtraction, round to the fewest:

- (A) Sig figs
- (B) Decimal places
- (C) Digits
- (D) Non-zero digits

Answer: _____

LESSON 3: DENSITY

Objective:

- Calculate density using a numerical set-up
- Solve for mass or volume given density

Density is the amount of matter in a given amount of space — the ratio of mass to volume: density = mass ÷ volume. Density is an intensive property: a small piece and a large block of the same substance have the same density. Units are g/mL or g/cm³.

NUMERICAL SET-UP RULE: Always write the formula, substitute the numbers WITH units, and give the answer with units. Example: $d = 74.6 \text{ g} \div 27.6 \text{ cm}^3 = 2.70 \text{ g/cm}^3$.

Worked Example

A sample of aluminum has a mass of 74.6 g and a volume of 27.6 cm³. Calculate its density.

Step 1: Write the formula:

$$d = \frac{m}{V}$$

Step 2: Substitute with units:

$$d = \frac{74.6 \text{ g}}{27.6 \text{ cm}^3}$$

Step 3: Divide and give units:

$$d = 2.70 \text{ g/cm}^3$$

Answer: 2.70 g/cm³

Check Your Understanding

1. Density is calculated as:

- (A) mass × volume
- (B) mass ÷ volume
- (C) volume ÷ mass
- (D) mass + volume

Answer: _____

2. Calculate density if 12.50 g has a volume of 4.6 mL (correct sig figs):

- (A) 2.717 g/mL
- (B) 2.72 g/mL
- (C) 2.7 g/mL
- (D) 3 g/mL

Answer: _____

3. Is density extensive or intensive?

- (A) Extensive
- (B) Intensive
- (C) Chemical
- (D) Neither

Answer: _____

4. What are the units of density?

- (A) g/mL or g/cm³
- (B) mL only
- (C) g only
- (D) cm³ only

Answer: _____

LESSON 4: SCIENTIFIC NOTATION

Objective:

- Convert standard notation to scientific notation
- Convert scientific notation to standard notation

Scientific notation writes numbers as a decimal between 1 and 10 multiplied by a power of ten: $N \times 10^n$. Converting from standard to scientific: move the decimal point so one non-zero digit is to the left; count the moves — left moves are positive exponents, right moves are negative.

Examples: 1,902,300,000 $\rightarrow 1.9023 \times 10^9$ (decimal moved 9 places left). 0.0000025 $\rightarrow 2.5 \times 10^{-6}$ (moved 6 places right).

Worked Example

Write 0.003080 in scientific notation.

Step 1: Move the decimal so one non-zero digit (3) is to the left — moved 3 places right.

Step 2: Right moves = negative exponent:

$$0.003080 = 3.080 \times 10^{-3}$$

Answer: 3.080×10^{-3}

Check Your Understanding

1. Write 1,902,300,000 in scientific notation:

- (A) 1.9023×10^9
- (B) 19.023×10^8
- (C) 1.9023×10^{-9}
- (D) 1.9×10^9

Answer: _____

2. Write 0.0000025 in scientific notation:

- (A) 2.5×10^{-6}
- (B) 2.5×10^6
- (C) 25×10^{-7}
- (D) 2.5×10^{-5}

Answer: _____

3. The exponent in scientific notation is positive when the decimal moves:

- (A) Right (number < 1)
- (B) Left (number > 10)
- (C) Never
- (D) Only for zeros

Answer: _____

4. Write 4.5×10^{-3} in standard notation:

- (A) 4500
- (B) 0.0045
- (C) 0.045
- (D) 450

Answer: _____

LESSON 5: DIMENSIONAL ANALYSIS

Objective:

- Write conversion factors as ratios
- Convert units by canceling units

Dimensional analysis (unit conversion) uses conversion factors written as ratios to cancel unwanted units. A conversion factor is a number or ratio (in fraction form) used to convert from one unit to another. The given units cancel out, leaving the desired target units.

Example: $12.0 \text{ oz} \times (29.57 \text{ mL} / 1 \text{ oz}) = 355 \text{ mL}$ — the oz units cancel, leaving mL. Always set up so the unit you want to get rid of is in the denominator.

Worked Example

A 197-lb weight lifter wants to compete in a class limited to 90 kg. Is he light enough? (1 kg = 2.205 lb)

Step 1: Write the conversion factor with the desired unit (kg) on top:

$$\frac{1 \text{ kg}}{2.205 \text{ lb}}$$

Step 2: Multiply — the lb units cancel:

$$197 \text{ lb} \times \frac{1 \text{ kg}}{2.205 \text{ lb}} = 89.3 \text{ kg}$$

Step 3: Compare: $89.3 \text{ kg} < 90 \text{ kg}$ — he qualifies.

Answer: Yes — 89.3 kg, under the 90 kg limit.

Check Your Understanding

1. How many mL are in a 12.0-oz can? (1 oz = 29.57 mL)

- (A) 355 mL
- (B) 405 mL
- (C) 295 mL
- (D) 0.406 mL

Answer: _____

2. Convert 2.0 L to milliliters:

- (A) 20 mL
- (B) 200 mL
- (C) 2000 mL
- (D) 0.002 mL

Answer: _____

3. A conversion factor is best written as:

- (A) A ratio in fraction form
- (B) A single number
- (C) A word
- (D) A graph

Answer: _____

4. Convert 250 mL to liters:

- (A) 2.5 L
- (B) 0.25 L
- (C) 25 L
- (D) 0.025 L

Answer: _____

Regents Practice — Chapter 1

Answer in the style of the NY State Regents exam.

1.) Which mass measurement contains four significant figures?

- (1) 3200 g
- (2) 0.0032 g
- (3) 320.0 g
- (4) 0.0320 g

2.) During a laboratory experiment, a sample of aluminum is found to have a mass of 12.50 grams and a volume of 4.6 milliliters. What is the density, expressed to the correct number of significant figures?

- (1) 2.717 g/mL
- (2) 2.72 g/mL
- (3) 3 g/mL
- (4) 2.7 g/mL

3.) Which quantity of heat is equal to 200. joules in kilojoules?

- (1) 20.0 kJ
- (2) 2.00 kJ
- (3) 0.200 kJ
- (4) 0.0200 kJ

4.) A student measures the mass of a sample as 24.56 g. The accepted value is 24.60 g. The student's measurement is best described as:

- (1) Accurate only
- (2) Precise only
- (3) Both accurate and precise
- (4) Neither

5.) Express 1,902,300,000 in scientific notation:

- (1) 1.9023×10^9
- (2) 19.023×10^8
- (3) 1.9023×10^{-9}
- (4) 1.9023×10^6

6.) The diagram shows a 100-milliliter graduated cylinder with the meniscus between the 35.0 and 36.0 mL marks. What is the correct reading?

- (1) 35.0 mL
- (2) 35.5 mL
- (3) 44.0 mL
- (4) 45.0 mL

7.) Expressed to the correct number of significant figures, what is the sum of 210.10 g + 235.100 g?

- (1) 445.2 g
- (2) 445.20 g
- (3) 445.200 g
- (4) 445 g

8.) Which measurement is a QUALITATIVE observation?

- (1) The mass is 5.0 g
- (2) The solution turned blue
- (3) The temperature is 298 K
- (4) The volume is 25.0 mL

Answer Key

Answers to all lesson practice and Regents practice questions.